

lab04_Goes

1

```
library(readxl)
```

Warning: pacote 'readxl' foi compilado no R versão 4.5.3

```
library(tidyverse)
```

Warning: pacote 'tidyverse' foi compilado no R versão 4.5.3

Warning: pacote 'ggplot2' foi compilado no R versão 4.5.3

Warning: pacote 'tibble' foi compilado no R versão 4.5.3

Warning: pacote 'tidyr' foi compilado no R versão 4.5.3

Warning: pacote 'readr' foi compilado no R versão 4.5.3

Warning: pacote 'purrr' foi compilado no R versão 4.5.3

Warning: pacote 'dplyr' foi compilado no R versão 4.5.3

Warning: pacote 'stringr' foi compilado no R versão 4.5.3

Warning: pacote 'forcats' foi compilado no R versão 4.5.3

Warning: pacote 'lubridate' foi compilado no R versão 4.5.3

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

2

```
fname <- "WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

3

```
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 264 x 31
```

	`Country Code` <chr>	`Short Name` <chr>	`Table Name` <chr>	`Long Name` <chr>	`2-alpha code` <chr>
1	ABW	Aruba	Aruba	Aruba	AW
2	AFE	Africa Eastern and So~	Africa East~	Africa Eas~	ZH
3	AFG	Afghanistan	Afghanistan	Islamic St~	AF
4	AFW	Africa Western and Ce~	Africa West~	Africa Wes~	ZI
5	AGO	Angola	Angola	People's R~	AO
6	ALB	Albania	Albania	Republic o~	AL
7	AND	Andorra	Andorra	Principali~	AD
8	ARB	Arab World	Arab World	Arab World	1A
9	ARE	United Arab Emirates	United Arab~	United Ara~	AE
10	ARG	Argentina	Argentina	Argentine ~	AR

```
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

4

```
países_escolhidos <- c("Brazil", "Argentina", "Chile", "Uruguay", "Paraguay")
countries <- countries %>%
  filter(`Table Name` %in% países_escolhidos)
countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name` `Table Name` `Long Name`      `2-alpha code`
  <chr>          <chr>        <chr>        <chr>          <chr>
1 ARG          Argentina    Argentina    Argentine Republic AR
2 BRA          Brazil      Brazil      Federative Republic of Brazil BR
3 CHL          Chile       Chile       Republic of Chile   CL
4 PRY          Paraguay    Paraguay    Republic of Paraguay PY
5 URY          Uruguay     Uruguay     Oriental Republic of Uruguay UY
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

5

```
WDI <- read_excel(fname, sheet = "Data")
WDI
```

```
# A tibble: 396,970 x 70
  `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
```

```

      <chr>      <chr>      <chr>      <chr>      <dbl> <dbl>
1 Africa Easter~ AFE      Access to clean~ EG.CFT.ACCS.ZS      NA      NA
2 Africa Easter~ AFE      Access to clean~ EG.CFT.ACCS.RU.~      NA      NA
3 Africa Easter~ AFE      Access to clean~ EG.CFT.ACCS.UR.~      NA      NA
4 Africa Easter~ AFE      Access to elect~ EG.ELC.ACCS.ZS      NA      NA
5 Africa Easter~ AFE      Access to elect~ EG.ELC.ACCS.RU.~      NA      NA
6 Africa Easter~ AFE      Access to elect~ EG.ELC.ACCS.UR.~      NA      NA
7 Africa Easter~ AFE      Account ownersh~ FX.OWN.TOTL.ZS      NA      NA
8 Africa Easter~ AFE      Account ownersh~ FX.OWN.TOTL.FE.~      NA      NA
9 Africa Easter~ AFE      Account ownersh~ FX.OWN.TOTL.MA.~      NA      NA
10 Africa Easter~ AFE      Account ownersh~ FX.OWN.TOTL.OL.~      NA      NA
# i 396,960 more rows
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, ...

```

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```

WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
WDI

```

```

# A tibble: 396,970 x 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 AFE           EG.CFT.ACCS.ZS      NA     NA     NA     NA     NA     NA
2 AFE           EG.CFT.ACCS.RU.ZS    NA     NA     NA     NA     NA     NA
3 AFE           EG.CFT.ACCS.UR.ZS    NA     NA     NA     NA     NA     NA
4 AFE           EG.ELC.ACCS.ZS      NA     NA     NA     NA     NA     NA
5 AFE           EG.ELC.ACCS.RU.ZS    NA     NA     NA     NA     NA     NA
6 AFE           EG.ELC.ACCS.UR.ZS    NA     NA     NA     NA     NA     NA
7 AFE           FX.OWN.TOTL.ZS      NA     NA     NA     NA     NA     NA
8 AFE           FX.OWN.TOTL.FE.ZS    NA     NA     NA     NA     NA     NA
9 AFE           FX.OWN.TOTL.MA.ZS    NA     NA     NA     NA     NA     NA
10 AFE          FX.OWN.TOTL.OL.ZS    NA     NA     NA     NA     NA     NA
# i 396,960 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,

```

```
# `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
# `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
# `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
# `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...
```

7

```
WDI <- WDI %>%
  pivot_longer(
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "VALUE"
  )
WDI
```

```
# A tibble: 23,421,230 x 4
  `Country Code` `Indicator Code` YEAR  VALUE
  <chr>          <chr>          <chr> <dbl>
1 AFE           EG.CFT.ACCS.ZS  1960    NA
2 AFE           EG.CFT.ACCS.ZS  1961    NA
3 AFE           EG.CFT.ACCS.ZS  1962    NA
4 AFE           EG.CFT.ACCS.ZS  1963    NA
5 AFE           EG.CFT.ACCS.ZS  1964    NA
6 AFE           EG.CFT.ACCS.ZS  1965    NA
7 AFE           EG.CFT.ACCS.ZS  1966    NA
8 AFE           EG.CFT.ACCS.ZS  1967    NA
9 AFE           EG.CFT.ACCS.ZS  1968    NA
10 AFE          EG.CFT.ACCS.ZS  1969    NA
# i 23,421,220 more rows
```

8

```
WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))
is.integer(WDI$YEAR)
```

```
[1] TRUE
```

WDI

```
# A tibble: 23,421,230 x 4
  `Country Code` `Indicator Code` YEAR VALUE
  <chr>          <chr>          <int> <dbl>
1 AFE            EG.CFT.ACCS.ZS    1960    NA
2 AFE            EG.CFT.ACCS.ZS    1961    NA
3 AFE            EG.CFT.ACCS.ZS    1962    NA
4 AFE            EG.CFT.ACCS.ZS    1963    NA
5 AFE            EG.CFT.ACCS.ZS    1964    NA
6 AFE            EG.CFT.ACCS.ZS    1965    NA
7 AFE            EG.CFT.ACCS.ZS    1966    NA
8 AFE            EG.CFT.ACCS.ZS    1967    NA
9 AFE            EG.CFT.ACCS.ZS    1968    NA
10 AFE           EG.CFT.ACCS.ZS    1969    NA
# i 23,421,220 more rows
```

9

```
WDI <- WDI %>%
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%
  inner_join(
    countries %>% select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) %>%
  rename(SP.DYN.CBRT.IN = VALUE) %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
WDI
```

```
# A tibble: 295 x 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>          <int>      <dbl>
1 ARG            Argentina    1960      24.2
2 ARG            Argentina    1961      23.9
3 ARG            Argentina    1962      23.8
4 ARG            Argentina    1963      23.5
5 ARG            Argentina    1964      23.2
6 ARG            Argentina    1965      22.6
7 ARG            Argentina    1966      22.3
8 ARG            Argentina    1967      22.4
```

```

 9 ARG      Argentina      1968      22.7
10 ARG      Argentina      1969      23.0
# i 285 more rows

```

10

```

WDI <- WDI %>%
  arrange(`Short Name`, YEAR)
WDI

```

```

# A tibble: 295 x 4
  `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
  <chr>          <chr>        <int>      <dbl>
1 ARG      Argentina      1960      24.2
2 ARG      Argentina      1961      23.9
3 ARG      Argentina      1962      23.8
4 ARG      Argentina      1963      23.5
5 ARG      Argentina      1964      23.2
6 ARG      Argentina      1965      22.6
7 ARG      Argentina      1966      22.3
8 ARG      Argentina      1967      22.4
9 ARG      Argentina      1968      22.7
10 ARG     Argentina      1969      23.0
# i 285 more rows

```

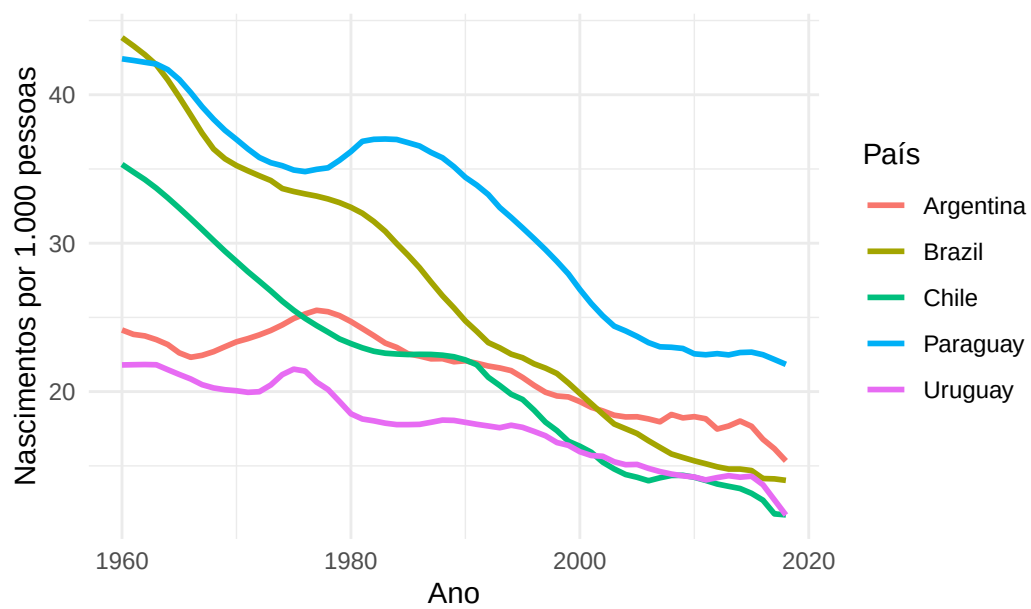
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```

ggplot(WDI, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Taxa Bruta de Natalidade (1960 - 2018)",
    x = "Ano",
    y = "Nascimentos por 1.000 pessoas",
    color = "País"
  ) +
  theme_minimal()

```

Taxa Bruta de Natalidade (1960 – 2018)



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-08 19:39:02 -03"
```